

Novel Chiral Catalysts for Asymmetric Addition of Terminal Alkynes to Aldehydes

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Abstract: There is growing interest in enantioselective synthesis of optically active propargylic alcohols. One of the most effective and straightforward ways for their synthesis is asymmetric addition of terminal alkynes to aldehydes. In the past several years, various combinations of novel chiral ligands and transition metal sources have been employed as catalysts in this asymmetric alkynylation reaction. In this review, we highlight the recent advances in this field and try to compare the methods available for this transformation.

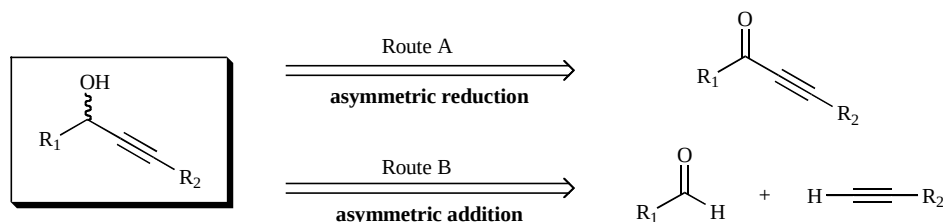
1. INTRODUCTION

Optically active propargylic alcohols are versatile precursors for the synthesis of many natural products and pharmaceuticals [1]. Although the most common approach to these compounds is the direct reduction of alkynyl ketones (Route A, Scheme 1), the alkynylation of aldehydes by organometallic reagents has become the preferred way of synthesizing the useful chiral propargylic alcohols (Route B, Scheme 1). The advantage of the latter is that the asymmetric alkynylzinc addition to carbonyl compounds can simultaneously form a new C-C bond and a stereogenic center in one step. Thus, the development of various chiral ligands as catalysts for the enantioselective alkynylations has gained considerable significance during the past years [2]. Here, we review recent developments in chiral catalysts for asymmetric alkynylations of aldehydes, with a focus on the different central metals, in both homogeneous or heterogeneous catalysis.

has been made since in this transformation using various zinc or other metal reagents. The following work will be presented in view of different central metals used.

2.1. Alkylzinc-Mediated Asymmetric Alkynylations of Aldehydes

There are many reports, spanning several decades, about asymmetric addition of organozinc reagents to aldehydes [2a]. In contrast, the efficient catalytic systems for the asymmetric addition of terminal alkynes to aldehydes have been achieved only in the past several years and remained less developed [4]. Among the examples in alkylzinc-mediated asymmetric alkynylations of aldehydes, chiral N,O-ligands have been amongst the most attractive. The proline-derived bimetallic catalyst system **1** developed by Trost and co-workers has proved a universal and efficient



Scheme 1. Common approaches to prepare optically active propargylic alcohols.

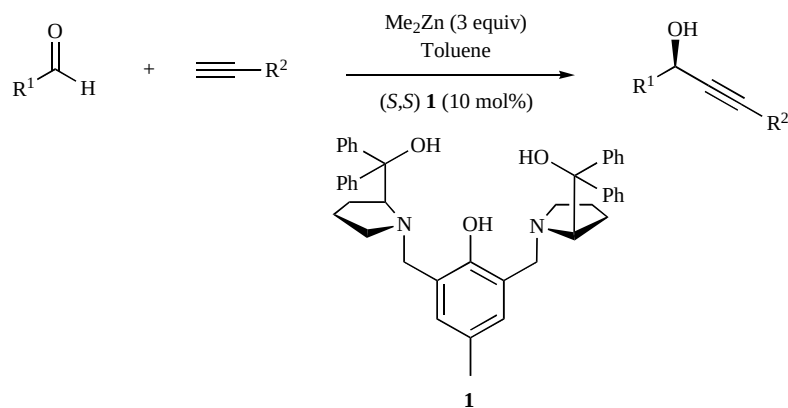
2. HOMOGENEOUS ENANTIOSELECTIVE ALKYNYLATION OF ALDEHYDES

In 1979, Mukaiyama and co-workers reported the pioneering work on the asymmetric version of asymmetric addition of a terminal alkyne to an aldehyde [3]. During their research, ⁿBuLi was used to deprotonate trimethylsilylacetylene and 4 equivalents of ligand were added. Great progress

catalyst for many enantioselective transformations [5]. Recently, this system also showed excellent chemical activities and enantioselectivities (up to 99% ee) for the asymmetric alkynylations of unsaturated aldehydes (Scheme 2) [6].

Wolf *et al.* reported a highly efficient asymmetric addition of phenylacetylene or aliphatic alkynes to both aromatic and aliphatic aldehydes in the presence of novel C₂-symmetric bisoxazolidines **2** [7]. However, under the same conditions, (1R, 2S)-cis-1-amino-2-indanol-derived oxazolidine **3** gave only 76% yield and 17% ee (Fig. 1). That C₂ or C₃-symmetric ligands gave better results than C₁-symmetric ligands can be easily rationalized [8].

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Scheme 2.

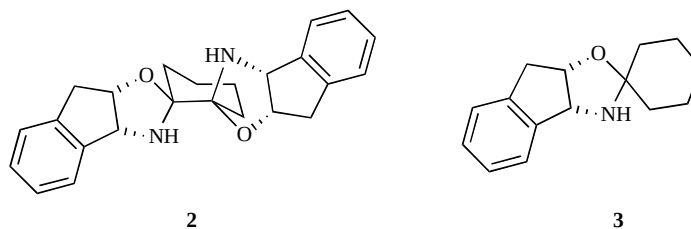
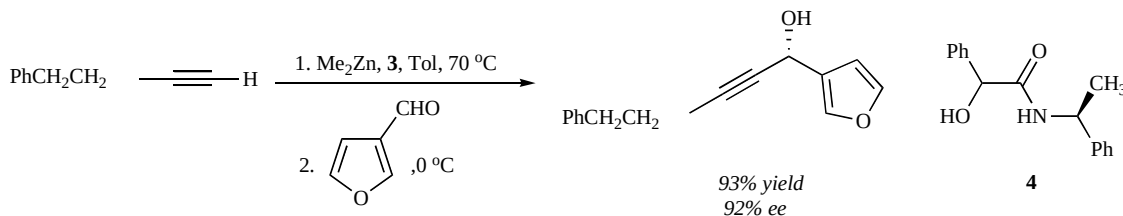


Fig. (1).



Scheme 3.

Blay and Pedro have reported that a one-step prepared (*S,S*)-mandelamide **4** could catalyze the asymmetric additions of aryl- and alkynylzinc reagents to heteroaromatic aldehydes with good yields (up to 94%) and good enantioselectivities (up to 92% ee) (Scheme 3) [9]. It is noteworthy that in order to improve the enantioselectivity of the reaction, the authors performed the alkynylzinc reagent prior to the addition of aldehyde. Effectively, it was necessary to heat at $70^\circ C$ phenylacetylene with dimethylzinc in toluene in the presence of **4** and then to add benzaldehyde at $0^\circ C$, which is different from the procedure described by Pu: ligand is added into the system after the formation of alkynylzinc reagent [4n].

Chiral ligands bearing a binaphthyl framework, such as BINOL and BINAP, have gained great attention in recent years for their excellent performance in a variety of catalytic asymmetric reactions [10]. Considering that the partially hydrogenated binaphthyl scaffold (H_8 -binaphthyl) could show positive effects in the catalytic reactions due to its more rigid structure [11], Lu and Chan have synthesized second generation chiral N,O-ligand **5** and applied it to asymmetric addition of alkynylzinc or arylzinc to aldehydes, affording the corresponding chiral propargylic alcohols and diarylmethanols in high yields and good enantioselectivities (Fig. 2) [12].

Recently, Wang and co-workers also prepared a new chiral N,O-ligand **6** from readily available L-(+)-methionine and used it in different types of asymmetric addition reactions [13]. In the presence of 10 mol% of DiMPEG [dimethoxypoly(ethylene glycol), MW 2000], [14] the chiral propargylic alcohols were obtained in good yields and moderate enantioselectivities (71-85% ee) (Fig. 2).

The enantioselective alkynylation of aldehydes mediated by chiral macrocycle **7** was achieved by Pu and co-workers [15]. Two equivalents of dimethylzinc and phenylacetylene in the presence of 20 mol% of ligand furnished the asymmetric addition on linear or branched aliphatic aldehydes and vinyl aldehydes in 89-96% ee (Fig. 2).

Mao and Wan have found that ephedrine-derived sulfamide-amine alcohol **8** was an effective catalyst for the asymmetric phenylacetylene addition to aldehydes at room temperature, affording the desired adduct in high yields (up to 99%) and good enantioselectivities (up to 84% ee) [16]. The weakly coordinative sulfonamide moiety of the ligand plays an important role in the acceleration and stereocontrol of these asymmetric transformations. Li and Wan (from the same research group) further made the catalytic alkynylzinc addition in the presence of **9** as the ligand. The improved results they obtained -99% yields and with 92% ee- sug-

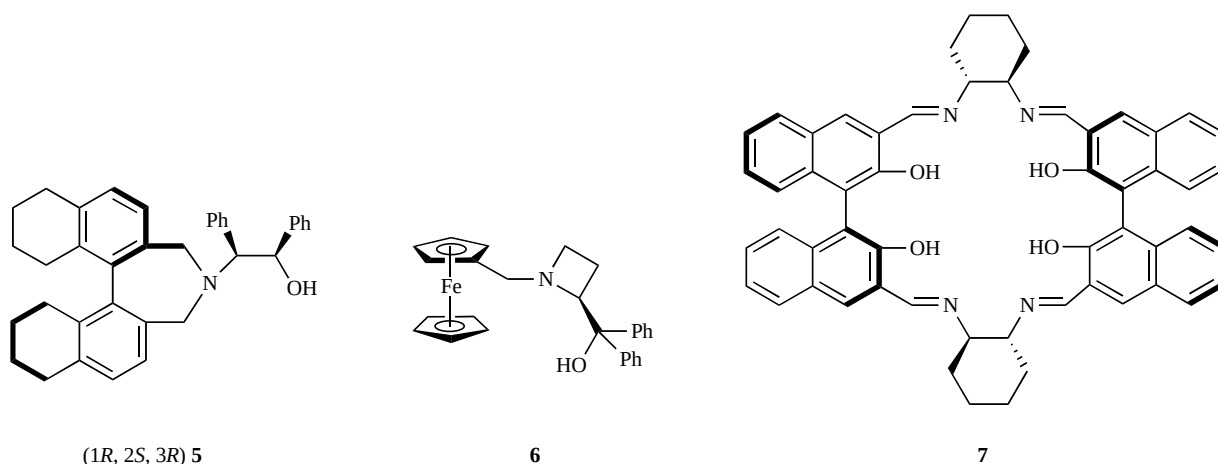


Fig. (2).

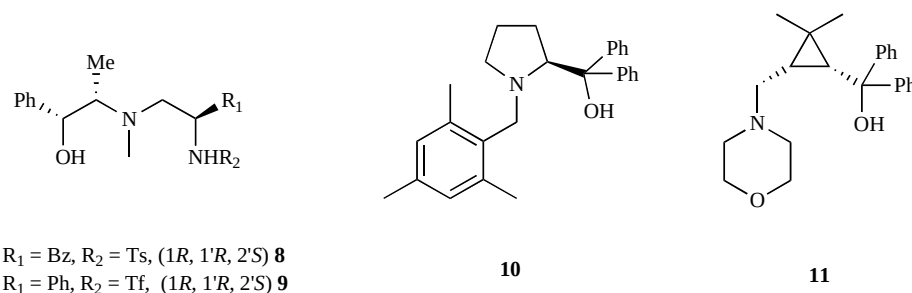


Fig. (3).

gested that the catalytic efficiency depends on the electronic effect and steric effect of the ligands (Fig. 3) [17].

Proline-derived tertiary amino alcohols have proved effective catalysts in asymmetric addition of diethylzinc to aldehydes [18]. However, they were not attractive in asymmetric alkynylation of aldehydes [4h]. Mao and Zhang recently surveyed a series of N,O-ligands based on proline. Optimal enantioselectivity (83% ee) was obtained in the presence of **10**. Meanwhile, they supported their results using theoretical calculations (Fig. 3) [19].

The novel chiral N,O-ligand **11** was used by Liu and Wang for alkynylzinc additions [20]. They found that the enantiomerically pure cyclopropane-based amino alcohol **11** catalyzed the reaction of phenylacetylene (2.6 equiv.) with aldehydes in the presence of Me₂Zn (2.4 equiv.) and DiMPEG (10 mol%) in toluene to give the corresponding chiral propargylic alcohols in high yields (up to 95%) and high enantioselectivities (up to 95% ee) (Fig. 3).

2.2. Ti(O^{*i*}Pr)₄-Promoted Alkynylation of Aldehydes

The pioneering work on titanium-mediated reactions was independently reported by Pu [4n] and Chan [4e]. The successful catalytic system combined BINOL and Ti(O^{*i*}Pr)₄, which proved effective in enantioselective alkynylation of various aldehydes, including aromatic, aliphatic and α,β-unsaturated aldehydes. Further structural modifications on the BINOL ligand were also reported for improved performance in these asymmetric transformations [40].

C₃ symmetry is intriguing in chemistry, and many C₃-symmetric compounds have been successfully applied to

asymmetric catalysis [21]. Du and Xu reported that new C₃-symmetric chiral tris(β-hydroxy amide) ligand **10** from (1*R*, 2*S*)-(-)-2-amino-1,2-diphenylethanol for enantioselective alkynylation of benzaldehyde in the presence of Ti(O^{*i*}Pr)₄, affording the desired product in 84% yield with 87% ee [22]. Under the same conditions, C₂- and C₁-symmetric ligands **11** and **12** were synthesized and applied in the catalytic reactions. It is obvious to see that decreased enantioselectivities were obtained, which demonstrates the efficiency of C₃-symmetric ligand (Fig. 4).

However, it is a dilemma that C₂ or C₃-symmetric ligands definitely have higher synthetic cost and difficulty than C₁-symmetric ligand. Therefore, to develop easily accessible and operationally simple ligands remains a challenge. Enlightened by design of bisoxazolidines **2**, [7] Mao and Zhang have prepared a series of C₁-symmetric oxazolidines from readily available (1*R*, 2*S*)-cis-1-amino-2-indanol in good yields. When oxazolidine **13** was combined with titanium (IV) isopropoxide, aromatic aldehydes were alkynylated in up to 98% yield and up to 95% ee as shown in Table 1 [23].

It is possible that the titanium salt acts not only as a Lewis acid, but participates into the coordination, which may generate a new active asymmetric catalyst. Recently, Mao surveyed many readily available chiral amino alcohols for the titanium-mediated asymmetric alkynylation of aldehydes: the results showed that unmodified (1*R*, 2*S*)-(-)-2-amino-1,2-diphenylethanol **14** in combination with Ti(O^{*i*}Pr)₄ could catalyze the reaction of an alkynylzinc reagent with various aldehydes to generate chiral propargylic alcohols with moderate to good yields and enantioselectivities [24].

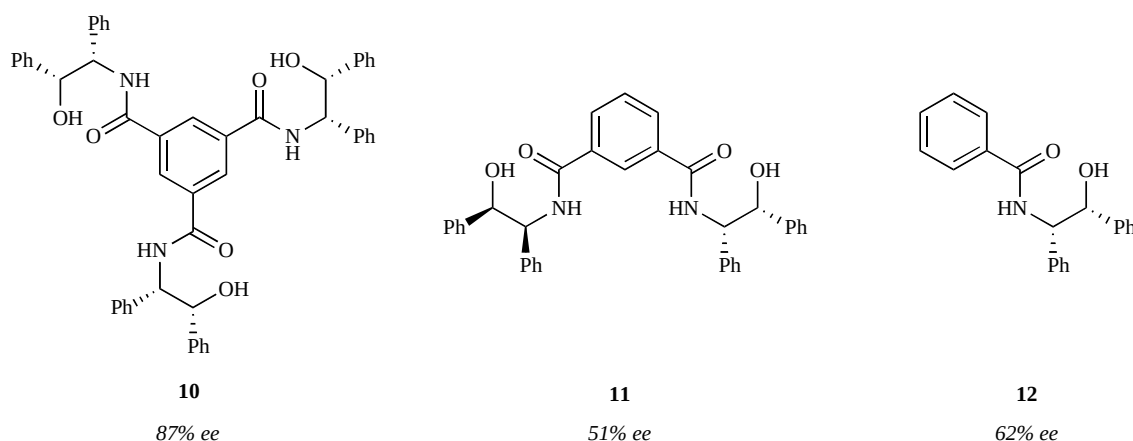
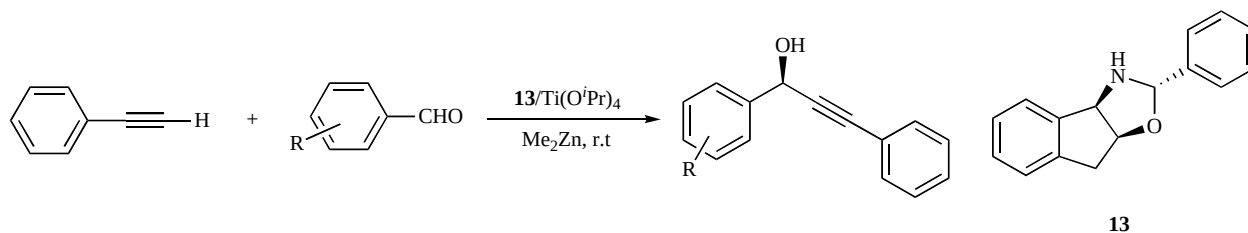


Fig. (4).

Table 1. Enantioselective Alkynylation of Phenylacetylene to Various Aldehydes with **13**^a

Entry	Aldehyde	Yield (%) ^b	ee (%) ^c
1	benzaldehyde	98	90
2	2-anisaldehyde	91	92
3	4-anisaldehyde	91	95
4	4-tolualdehyde	87	90
5	2-chloroaldehyde	96	93
6	3-chloroaldehyde	97	91
7	α -naphthaldehyde	96	90
8	β -naphthaldehyde	97	93
9	2,3-dimethoxybenzaldehyde	91	90
10	2-furaldehyde	91	90
11	hydrocinnamaldehyde	81	77

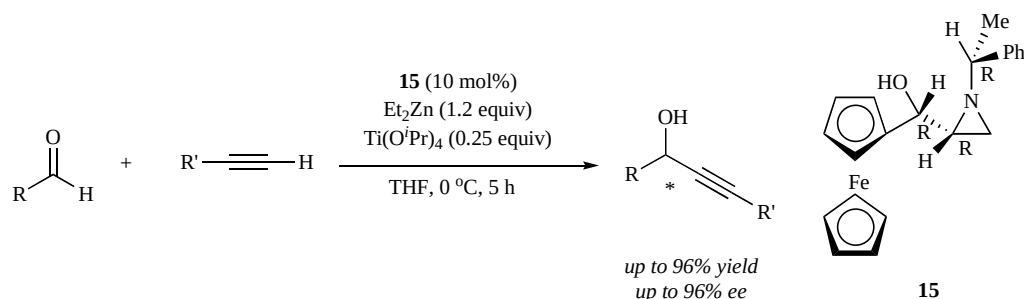
^a All the reactions were processed under argon at room temperature for 20h. phenylacetylene/Et₂Zn/benzaldehyde/**13**/Ti(O^{*i*}Pr)₄ = 2 : 2 : 0.5 : 0.1 : 0.2. ^b Isolated yield. ^c The enantiomeric excess was determined by chiral HPLC analysis of the corresponding products on a Chiralcel OD-H column.

Recently, Dogan and co-workers prepared the optically active ferrocenyl-substituted aziridinylmethanol **15** and studied its use in asymmetric alkynylation of aldehydes, affording the corresponding propargylic alcohols in up to 96% yield and 96% ee (Scheme 4) [25].

2.3. Zn(OTf)₂-Mediated Enantioselective Alkynylations of Aldehydes

Carreira and co-workers have reported two efficient systems using stoichiometric or catalytic amount of N-

methylephedrine **16** and Zn(OTf)₂ to promote the reaction of terminal alkynes with aldehydes to give high enantioselectivities (up to 99% ee) [3b]. This catalyst system was more suitable for aliphatic aldehydes, due to the Cannizzaro reaction. Based on this finding, Jiang and Tomioka probed similar systems based on ligand **17** and **18** as efficient catalysts in the asymmetric alkynylations [26, 27]. Tyrrell focused on the effect of substituents upon the efficiency and enantioselectivity using a **16**/Zn(OTf)₂ system, with interesting results [28]. Adolfsson and co-workers disclosed that cinchona alka-



Scheme 4.

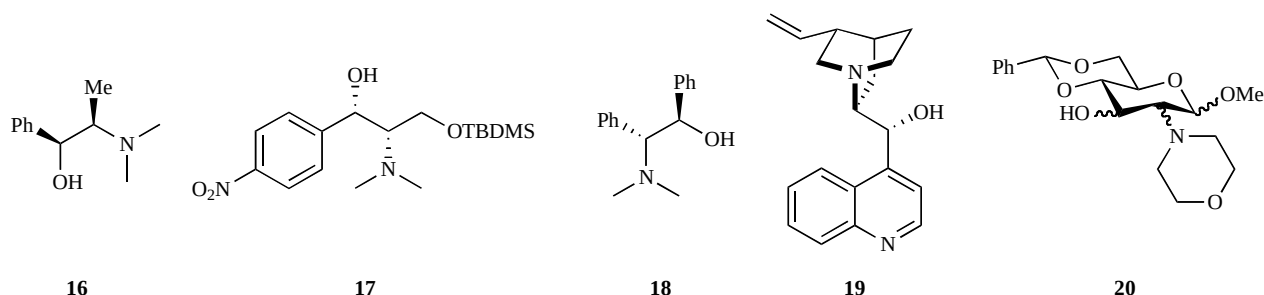
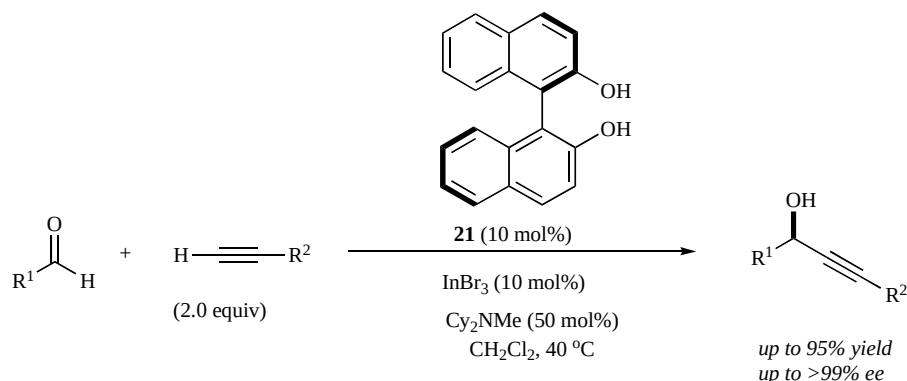


Fig. (5).



Scheme 5.

lolid **19** in combination with zinc triflate catalyze the addition of acetylenes to aldehydes at room temperature with ee up to 89% for aliphatic aldehydes [29]. Simultaneously, Emerson and Davis developed efficiently $Zn(OTf)_2$ -mediated enantioselective alkynylation of aldehydes in high ees in the presence of carbohydrate-derived amino alcohol **20** (Fig. 5) [30].

2.4. $InBr_3$ -Catalyzed Alkynylation of Aldehydes

In the past several years, bifunctional catalysis has become a successful concept in the design of novel efficient asymmetric catalysts for various enantioselective transformations [31]. The success of this catalysis is attributed to the dual activation of both substrates. Shibasaki and co-workers developed an efficient BINOL **21**/ $InBr_3$ system for asymmetric alkynylation of aldehydes, which has broad generality for both aromatic and aliphatic aldehydes, as well as phenylacetylene, alkenylacetylenes, and alkylacetylenes (Scheme 5) [32]. Meanwhile, when the reaction was carried out using nonenantiopure BINOL, rather strong positive non-linear effects were observed between the enantiomeric excess of BINOL and the desired product. In addition, they

BINOL and the desired product. In addition, they also found that using 2-methyl-3-butyne-2-ol as an ethyne equivalent donor, indium(III)-catalyzed asymmetric alkynylation of aryl, heteroaryl, alkyl and alkenyl aldehydes afforded desired products in moderate to good yields (up to 97%) and high enantioselectivities (up to 99% ee) [33].

2.5. Vanadium-Mediated Asymmetric Alkynyl Additions to Aldehydes

Gau and co-workers reported the first example of asymmetric alkynylation reactions employing a chiral oxovanadium(V) complex of tridentate Schiff base of β -amino alcohol (**22**) [34]. It is noteworthy that stereoselectivities of the propargyl alcohols are tunable through adjustments of steric sizes of substituents on Schiff base ligands and of the alkoxide ligand on the vanadium metal. Complex **22** catalyzed the reaction of phenylacetylene (3.0 equiv) with aromatic aldehydes in the presence of Et_2Zn (2.0 equiv) in the mixed solvent (toluene:dichloromethane = 2:1) with good to excellent enantioselectivities (73-99% ee) (Fig. 6) [35].

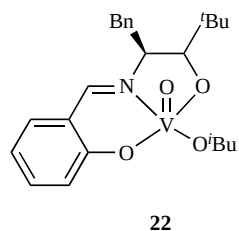


Fig. (6).

2.6. Copper-Promoted Enantioselective Addition of Terminal Alkynes to Aldehydes

Based on all of examples mentioned above, it appears that organozinc reagents were necessary to activate alkyne for the addition on aldehyde. However, Sawaura and co-worker found that the addition of terminal alkynes to aromatic aldehydes was carried out in presence of a Cu-phosphine **23** complex, yielding enantiomerically enriched propargyl alcohols with moderate ee values (Scheme 6) [36]. This is the first example of such an addition reaction without using organozinc reagents.

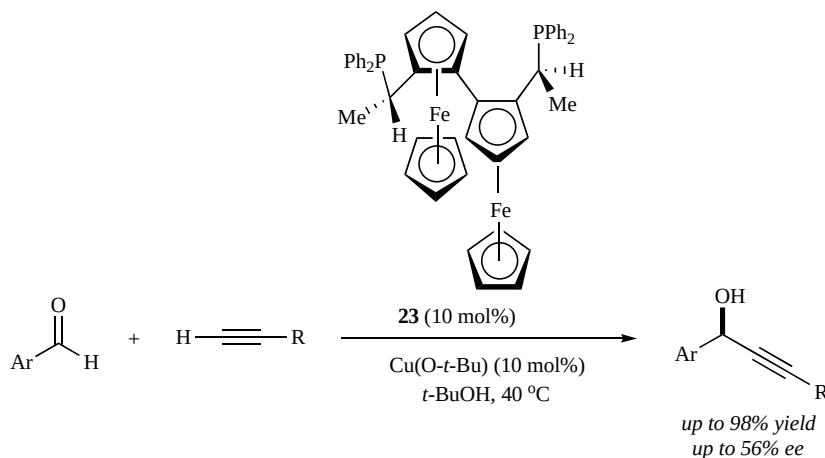
2.7. Asymmetric Alkynylations with Propiolate Nucleophiles

It is well established that γ -hydroxy- α,β -acetylenic esters containing three different functional groups are very important precursors in the synthesis of highly functionalized or-

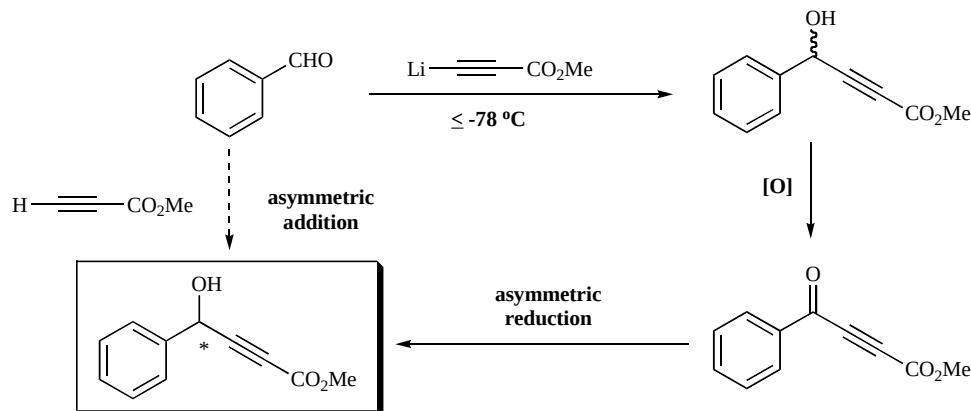
ganic molecules [37]. The traditional way to prepare these optically active esters is shown in Fig. (7) [38]. Obviously, this route is troublesome and tedious. Thus, a straightforward way was developed by the direct asymmetric addition of an alkyl propiolate to aldehydes. Although great efforts have been made in asymmetric alkynylations, relatively little attention has been paid to the enantioselective reactions of alkyne to aldehydes.

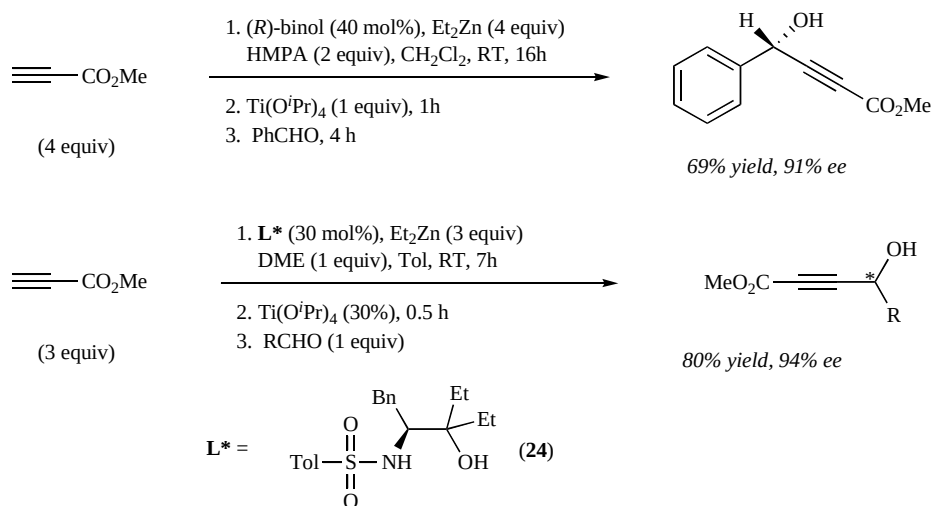
Recently, Pu and co-workers discovered that 2.0 equiv of hexamethylphosphoramide (HMPA) could accelerate the addition of methyl propiolate to various aldehydes with good yields and enantioselectivities in the presence of 40 mol% of BINOL as the chiral ligand [39]. Inspired by this finding, You reported that 5 mol% of *N*-methylimidazole (NMI) together with BINOL acting as dual catalytic system could catalyze enantioselective alkynylzinc addition to aromatic aldehydes with good results [40]. Wang and co-workers found that 1.0 equiv of 1,2-dimethoxyethane (DME) as an additive could facilitate the asymmetric addition of methyl propiolate to aldehydes with moderate to good ee values (up to 94%) and yields (up to 80%) using 30 mol% of β -sulfonamide alcohol **24** (Scheme 7) [41].

The proline-derived dinuclear zinc catalyst (1-Zn) developed by Trost has also proved an efficient catalyst for asymmetric addition of methyl propiolate to aldehydes [6]. Ligand **11** could catalytically synthesis of γ -hydroxy- α,β -acetylenic esters in good yields (up to 85%) and high enantioselectivities (up to 93% ee) [20]. Using 20 mol% of easily

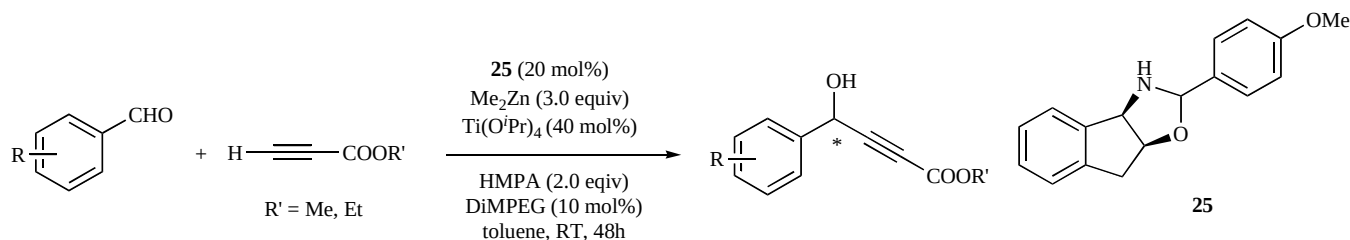


Scheme 6.

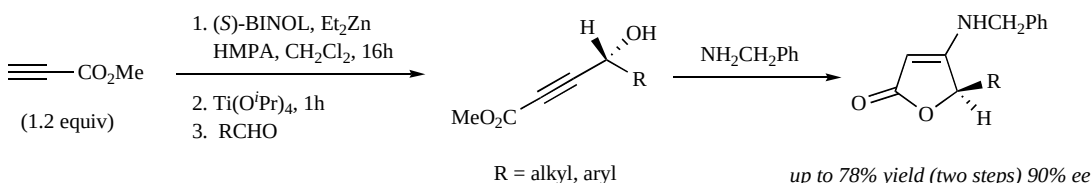
Fig. (7). The traditional way to prepare the optically active γ -hydroxy- α,β -acetylenic esters.



Scheme 7.



Scheme 8.



Scheme 9.

prepared oxazolidine **25** developed by Mao, the desired products were obtained in moderate to good yields (up to 81%) and enantioselectivities (up to 84% ee) in the presence of HMPA and DiMPEG (Scheme **8**) [42].

2(5*H*)-Furanone (or γ -butenolide) compounds have gained the pharmaceutical interest [43]. Pu discovered a convenient synthesis of optically active 4-amino-2(5*H*)-furanones by combining an asymmetric alkyne addition to aldehydes and a subsequent aliphatic amine addition. Using the BINOL/HMPA protocol, Pu got the desired products with high enantioselectivities (84–90% ee) (Scheme **9**) [44].

3. HETEROGENEOUS ENANTIOSELECTIVE ALKYNYLATION OF ALDEHYDES

The development of various chiral ligands as catalysts for the enantioselective alkyne addition has gained great significance during the past years. However, these homogeneous catalysts cannot be easily separated from products and hence present a serious obstacle for the synthetic applications. As a consequence of the increasing demand for efficient and environmentally friendly methods, heterogenization of homogeneous catalysts by immobilizing them with the liquid or solid supports has attracted a great deal of recent interest [45].

Although there have been several trials to immobilize homogeneous catalysts on solid supports, successful examples are still rare [46]. Therefore, there is urgent need to develop highly enantioselective and efficient chiral catalysts with broad substrate applicability and easy recyclability in this field.

Degni first reported asymmetric alkyne addition of benzaldehyde with chiral catalysts derived from *L*-prolinol immobilized on polymer fibers [47]. Although the high enantioselectivities (up to 91% ee) were achieved, high loading of catalyst (1.2 equivalent) and low conversions of the benzaldehyde (30–45%) confined their scale-up applications. Recently, Abdi and co-workers disclosed that a polymeric Zn(salen) could catalyze enantioselective addition of phenylacetylene to aldehydes and ketones. However, only 56% ee was obtained when benzaldehyde was used as the substrate [48]. Hui and Xu have developed a silica-immobilized titanium (IV) complex of β -hydroxyamide (**26**) in the enantioselective alkyne addition of benzaldehyde with moderate enantiomeric excesses (78% ee) (Fig. **8**) [49]. Wang reported asymmetric alkyne addition of ketones with polymer-supported chiral Schiff-base ligand (**27**) with good enantioselectivities

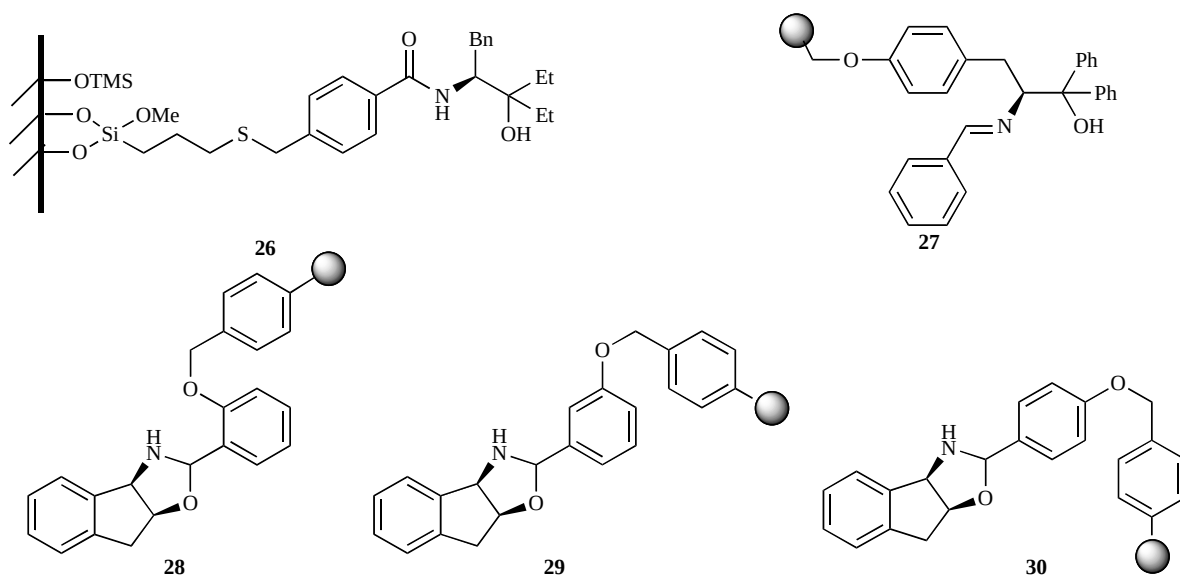


Fig. (8).

Table 2. Asymmetric Addition of Alkynes to Aromatic Aldehydes Promoted by Ligand **30**^a

Entry	Aldehyde	Alkyne	yield (%) ^b	ee (%) ^c
1	benzaldehyde	phenylacetylene	81	90
2	4-bromobenzaldehyde	phenylacetylene	84	92
3	4-bromobenzaldehyde	<i>p</i> -tolylacetylene	81	92
4	4-tolualdehyde	phenylacetylene	82	90
5	4-fluorobenzaldehyde	phenylacetylene	86	90
6	4-anisaldehyde	phenylacetylene	96	87
7	4-chlorobenzaldehyde	phenylacetylene	87	93
8	3-chlorobenzaldehyde	phenylacetylene	94	91
9	2-chlorobenzaldehyde	phenylacetylene	93	77
10	3-nitrobenzaldehyde	phenylacetylene	77	87
11 ^d	3-nitrobenzaldehyde	phenylacetylene	71	92
12	β -naphthaldehyde	phenylacetylene	88	94
13	β -naphthaldehyde	<i>p</i> -tolylacetylene	70	91
14	α -naphthaldehyde	phenylacetylene	82	86
15	2-thiophenealdehyde	phenylacetylene	98	95
16 ^e	2-thiophenealdehyde	phenylacetylene	84	88
17	2-thiophenealdehyde	<i>p</i> -tolylacetylene	92	95
18	2-furaldehyde	phenylacetylene	76	83
19	piperonaldehyde	phenylacetylene	86	92
20	piperonaldehyde	<i>p</i> -tolylacetylene	95	90
21	2,3-dimethoxybenzaldehyde	phenylacetylene	84	83

^a All the reactions were run in THF under argon atmosphere at room temperature for 20 h. $\text{Ti}(\text{O}^i\text{Pr})_4$ was freshly distilled. Phenylacetylene / Et_2Zn / benzaldehyde / ligand **30** / $\text{Ti}(\text{O}^i\text{Pr})_4$ = 2 : 2 : 0.5 : 0.14 : 0.28.

^b Isolated yield.

^c The enantiomeric excess was determined by chiral HPLC analysis of the corresponding products on a Chiralcel OD-H column.

^d **28** was used as the ligand.

^e Me_2Zn in place of Et_2Zn .

[50]. However, there is still large room for improvement in terms of catalyst enantioselectivity and reusability.

Recently, Mao discovered that the resin-supported oxazolidine-titanium complexes (**28-30**) could smoothly catalyze the asymmetric addition of terminal alkynes to aromatic aldehydes with high yields (up to 98 %) and enantioselectivities (up to 95% ee) (Table 2). In addition, this catalytic system was also suitable for asymmetric alkynylation of heteroaromatic aldehydes (up to 99% ee) and could be reused for 5 times after simple work-up (Fig. 8) [51].

4. CONCLUSIONS

The preparation of optically active chiral propargylic alcohols has achieved increasing attention in recent years, because they provide useful and versatile building blocks. Great efforts have been made to develop various efficient asymmetric catalysts. As a result of this research, many successful strategies and methods have led to big advances in this area. A remaining challenge is the catalytic asymmetric alkynylation of ketones [52], ketoesters [53] or imines [54], although several efficient catalytic systems have been developed in last five years. Comparatively, conjugate alkynylations are still under-developed [55]. It is anticipated that more and more chiral catalysts will be developed with increased substrate scope, reaction operability and catalytic efficiency, using lower catalyst loadings. The efficient reuse of such catalytic system is also likely to remain an important and challenging focus of research in the coming years.

ACKNOWLEDGEMENT

Financial support from the National Natural Science Foundation of China (No. 20802046) and the Key Laboratory of Organic Synthesis of Jiangsu Province. We also thank Dr. Marc Creus from University of Basel, Switzerland for English revision.

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